

WASP-12b

R-0563

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-12_170203 · created 2026-07-23T19:06:11+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2457788.69394 +/- 0.00035 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.153 +/- 0.044
Transit depth (Rp/Rs)^2	2.33 +/- 1.35 [%]
Orbital Inclination (inc)	82.9 +/- 2.82
Airmass coefficient 1 (a1)	0.39 +/- 0.46
Airmass coefficient 2 (a2)	1.02 +/- 1.57
Scatter in the residuals of the lightcurve fit is	179.24 %
Best Comparison Star	None
Optimal Aperture	1.22
Optimal Annulus	6.5
Transit Duration (day)	0.1269 +/- 0.0084

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.153 ± 0.044 vs 0.1178 (deviation 0.45σ)
Duration fit vs published	0.1269 d vs 0.125 d (deviation 0.13σ)
Mid-time O–C	1572.2 min
Depth SNR	2.0
Residual scatter	179.24 %

Light curve

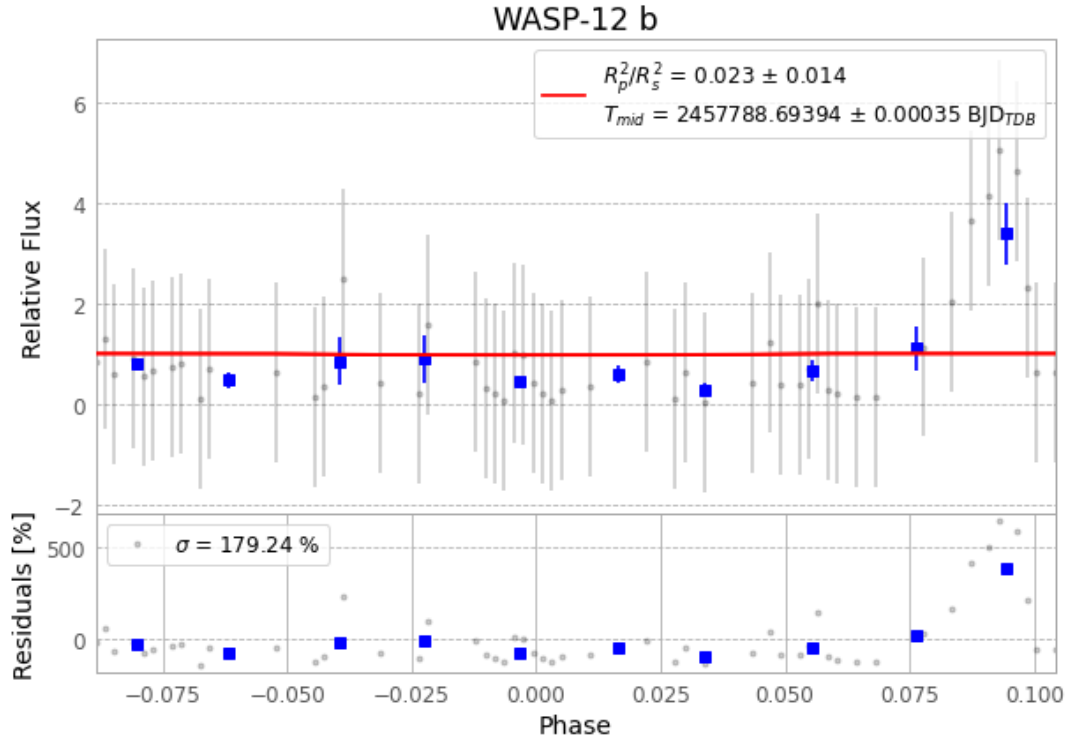


Figure 1 — FinalLightCurve_WASP-12 b_2017-02-03.png (SHA-256 211f81114160c620...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [577, 256], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 c2c83c0a1ba48805...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit b365494

Data provenance

102 FITS frames (67 MB), WASP-12170204021212.FITS → WASP-12170204071512.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-12-170203.log (572d9c20e484...), FinalLightCurve_WASP-12 b_2017-02-03.png (211f81114160...), FinalLightCurve_WASP-12 b_2017-02-03.csv (42686b8da819...)

Compute resources

Wall time 305.7 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-23 19:06Z.